

M1 Data Lakes / Data Intégration

Confluent/KsqlDB

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M1-DE3

Question: Décrire ce que fait la commande ligne par ligne :

```
CREATE STREAM transactions_stream ( transaction_id STRING, amount DOUBLE,  
transaction_type STRING ) WITH (  
KAFKA_TOPIC='transaction_log',VALUE_FORMAT='JSON' );
```

Cette commande permet de créer un stream avec *CREATE STREAM* avec le nom *transactions_stream*. On lui indique ensuite les champs de ce stream avec la key/value, *transaction_id STRING...*, puis on détermine le topic auquel il est associé *KAFKA_TOPIC='transaction_log'*, puis le format des messages *VALUE_FORMAT='JSON'*.

ksqldb1

Editor **Flow** Streams Tables Persistent queries Settings

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TRANSACTIONS_STREAM
STREAM

KSQL_PROCESSING_LOG
STREAM

Créez un nouveau stream pour les transactions importantes (dont le montant est supérieur à 700) *CREATE STREAM transactions_importantes_700 AS SELECT * FROM transactions_stream WHERE amount > 700;*

KSQL_PROCESSING_LOG
STREAM

TRANSACTIONS_STREAM
STREAM

CREATE-STREAM

...CTIONS_IMPORTANTES_700
STREAM

```

1 CREATE STREAM TRANSACTIONS_IMPORTANTES_700 WITH (CLEANUP_POLICY='delete',
2 KAFKA_TOPIC='TRANSACTIONS_IMPORTANTES_700', PARTITIONS=1, REPLICAS=1,
3 RETENTION_MS=604800000) AS SELECT *
4 FROM TRANSACTIONS_STREAM TRANSACTIONS_STREAM
5 WHERE (TRANSACTIONS_STREAM.AMOUNT > 700)
6 EMIT CHANGES;

```

• [Add query properties](#)

auto.offset.reset = Latest  

[+ Add another field](#)

Unable to run query.

Stop

Run query

 Invalid config variable in the WITH clause: CLEANUP_POLICY. The CLEANUP_POLICY config is automatically inferred based on the type of source (STREAM or TABLE). Users can't set the CLEANUP_POLICY config manually.

×

Total par type de transaction

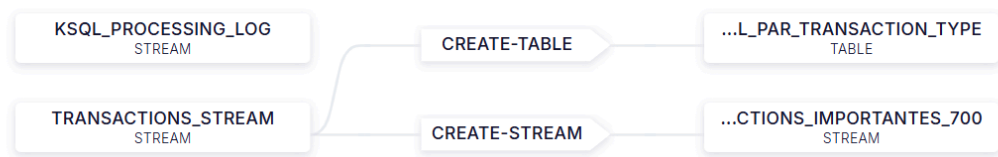
CREATE TABLE total_par_transaction_type AS SELECT transaction_type, SUM(amount) AS total_amount FROM transactions_stream GROUP BY transaction_type;

[Editor](#) [Flow](#) [Streams](#) [Tables](#) [Persistent queries](#) [Settings](#)

 Search tables

Table	Kafka topic			Data	
Name	Topic Name	Partitions	Replication	Key Format	Value Format
TOTAL_PAR_TRANSACTION_TYPE	TOTAL_PAR_TR...	3	1	KAFKA	JSON

Search



Exécutez le script python kafka_producer_transaction.py qui vous est fourni. Il va injecter quelques messages dans le topic d'entrée de votre pipeline transaction_log

Data structure
TABLE

Total messages
--

Messages/sec
--

Total message bytes
--

Message fields
The above statistics are computed for the query source.

- TRANSACTION_TYPE
- TOTAL_AMOUNT

Transaction messages:

- { "TRANSACTION_TYPE": "payment", "TOTAL_AMOUNT": 11479.399999999998 }
- { "TRANSACTION_TYPE": "payment", "TOTAL_AMOUNT": 11199.949999999997 }
- { "TRANSACTION_TYPE": "purchase", "TOTAL_AMOUNT": 12685.1 }
- { "TRANSACTION_TYPE": "refund", "TOTAL_AMOUNT": 10174.38 }

Questions:

**Parmi les transactions importantes, quel est le type le plus représenté?
Reportez votre requête SQL**

select TRANSACTION_TYPE, count() as NOMBRE_TRANSACTION from
TRANSACTIONS_IMPORTANTES_700 group by TRANSACTION_TYPE emit changes;*

The screenshot shows a data visualization tool interface. At the top, a SQL query is entered in a text area:

```
1 select TRANSACTION_TYPE, count(*) as NOMBRE_TRANSACTIONS from TRANSACTIONS_IMPORTANTES_700 group by TRANS
2
```

Below the query, there is a section for query properties. It includes a link "Add query properties", a field "auto.offset.reset" set to "Latest", and a "Running..." status with "Stop" and "Refresh" buttons. A "+Add another field" link is also present.

On the left, a sidebar shows the "table structure" (REAM) and "table messages" (messages/sec). The main area displays a message in a JSON format:

```
{ "TRANSACTION_TYPE": "withdrawal", "NOMBRE_TRANSACTIONS": 6 }
```

Le type le plus représenté est le withdrawal.

Quel est le schéma de la table TOTAL_PAR_TRANSACTION_TYPE (noms des colonnes et types) ?

Name Type
TRANSACTION_TYPE STRING
TOTAL_AMOUNT DOUBLE

Reportez une capture d'écran ou alors la requête SQL pour retrouver l'information ainsi que le résultat.

describe TOTAL_PAR_TRANSACTION_TYPE;

```
1 DESCRIBE total_par_transaction_type;
2
```

● [Add query properties](#)

auto.offset.reset = Latest 

+Add another field

Stop

Run query

```
1 {
2   "@type": "sourceDescription",
3   "statementText": "DESCRIBE total_par_transaction_type;",
4   "sourceDescription": {
5     "name": "TOTAL_PAR_TRANSACTION_TYPE",
6     "windowType": null,
7     "readQueries": [
8
9   ],
10  "writeQueries": [
11    {
12      "queryString": "CREATE TABLE TOTAL_PAR_TRANSACTION_TYPE WITH (CLEANUP_POLICY='compact',
KAFKA_TOPIC='TOTAL_PAR_TRANSACTION_TYPE', PARTITIONS=1, REPLICAS=1, RETENTION_MS=604800000) AS SELECT\n
TRANSACTIONS_STREAM.TRANSACTION_TYPE TRANSACTION_TYPE,\n SUM(TRANSACTIONS_STREAM.AMOUNT) TOTAL_AMOUNT\nFROM TRANSACTIONS_STREAM
TRANSACTIONS_STREAM\nGROUP BY TRANSACTION_TYPE\nEMIT CHANGES;",
13      "sinks": [
14        "TOTAL_PAR_TRANSACTION_TYPE"
15      ],
16      "sinkKafkaTopics": [
17        "TOTAL_PAR_TRANSACTION_TYPE"
18      ],
19      "id": "CTAS_TOTAL_PAR_TRANSACTION_TYPE_5",
20      "statusCount": {
21        "RUNNING": 1
22      },
23      "queryType": "PERSISTENT",
```

**Parmi les transactions importantes, reportez le montant moyen.
Reportez la requête SQL ou alors toute méthode utilisée.**

select avg(AMOUNT) as average from TRANSACTIONS_IMPORTANTES_700 emit changes;

1

```
select avg(AMOUNT) as average from TRANSACTIONS_IMPORTANTES_700 emit changes;
```

● [Add query properties](#)

auto.offset.reset

=

Latest

▼

🗑️

[+Add another field](#)

Running...

Stop

🔄

Data structure

STREAM

Total messages

--

Messages/sec

▶️ ⏸️

☰ ⏸️

▼

{ "AVERAGE": 855.3564285714285 }

Pour le stream TRANSACTIONS_IMPORTANTES_700, de combien est la période de rétention des données dans Kafka? Pourquoi cette période de rétention est-elle nécessaire?

La rétention est de 7j. Elle permet de conserver les données pendant un temps donné pour relancer le flux en cas de panne ou de relancement, limiter l'espace disque et de garantir que seules les données récentes sont disponibles pour les traitements temps réel.

TRANSACTIONS_IMPORTANTES_700

[Overview](#) [Messages](#) [Schema](#) [Configuration](#)

General settings

name	TRANSACTIONS_IMPORTANTES_700
partitions	3
min.insync.replicas	1
cleanup.policy	delete
retention.ms	604800000
max.message.bytes	1048588
retention.bytes	-1

[✎ Edit settings](#)

[☰ Show full config](#)

[!\[\]\(96cc62f861fdd6e50510c0224a756dff_img.jpg\) Delete topic](#)